

AgriFlow AI Master PRD

Project Title

Design and Implementation of an AI-Driven Agricultural Supply Chain and Produce Scheduling System with Intelligent Demand Forecasting.

Objective

Build a demo-ready final-year web application connecting Farmers, Buyers, Transporters, Warehouse Managers, and Administrators.

Technology Stack

Next.js 14, TypeScript, Tailwind CSS, Supabase (Postgres/Auth/Storage), Google Gemini, Leaflet, Recharts, Vercel.

Frozen Architecture

Keep database schema, authentication, role system, dashboard routing, shared dashboard shell, folder structure, and Supabase integration unless a genuine implementation issue requires change.

Core Principles

Every feature must work without Gemini. Gemini only explains, summarizes, recommends, and answers questions. Business logic is deterministic.

Modules

1. Authentication & Role Management
2. Farmer Module (crops, harvests, calendar)
3. Buyer Module (browse, order, track)
4. Warehouse Module (inventory, storage, stock movement)
5. Transport Module (deliveries, status, routes)
6. Scheduling Engine (rule-based)
7. Demand Forecasting (moving average, weighted moving average)
8. Notifications (rule-based)
9. AI Assistant (Gemini explanations only)
10. Reports & Analytics
11. Admin Dashboard

Algorithms

Scheduling: rule-based.

Forecasting: moving average and weighted moving average.

Spoilage: shelf-life rules.

Routing: nearest-distance calculation with Leaflet.

UI Requirements

Responsive SaaS dashboard, sidebar, navbar, cards, tables, charts, loading, skeletons, empty states, reusable components.

Code Standards

Strict TypeScript, Server Components by default, Server Actions where appropriate, reusable utilities/components, no duplicate business logic.

Workflow

Implement complete modules, run npm run lint and npm run build after each major module, fix all errors before continuing. Only stop for missing credentials or unrecoverable issues.

Definition of Done

Working authentication, CRUD, dashboards, scheduling, forecasting, reports, AI assistant, responsive UI, successful production build, ready for Vercel deployment.